

INSIGHTS FROM THE RETRIEVED DATA

Using the provided datasets and relational schema of TARGET's e-commerce operations in Brazil, I performed multiple SQL-based analyses to understand different aspects of the business, including customer distribution, purchasing behavior, sales trends, order fulfillment, revenue generation, freight costs, and payment preferences.

Through the INFORMATION_SCHEMA, I obtained an overall understanding of the database structure, including the base and junction entities, their columns and data types, primary keys, foreign keys, and the relationships between different entities.

1. Geospatial Analysis

TARGET's customers placed orders between **September 2016 and October 2018**, covering a wide range of cities and states across Brazil.

The `customers` table contains **99,441 records**, covering **4,119 unique cities across 27 states**. A significant proportion of orders originated from **São Paulo and Rio de Janeiro**, particularly from their respective capital cities.

The higher concentration of orders in these regions can potentially be associated with their large populations, strong economic activity, and high levels of urbanization. São Paulo, in particular, is one of Brazil's major economic and financial centers. Higher population density and purchasing power can naturally contribute to greater e-commerce demand.

Other possible contributing factors include the convenience of online shopping for urban consumers, stronger brand awareness, effective marketing activities, efficient distribution networks, and customer-oriented product offerings. However, these factors represent possible business explanations and would require additional data to establish direct causality.

2. Customer Lifecycle Value

By integrating data from the `orders` and `customers` tables, TARGET can evaluate customer purchasing behavior and potentially estimate the future value of customers based on their historical transactions.

Analysis of order placement times shows that customers frequently placed orders during the **afternoon, particularly between 1 PM and 6 PM**.

This pattern indicates that the afternoon may represent a potentially valuable period for customer engagement. TARGET could use this insight to experiment with targeted promotions, personalized offers, or marketing campaigns during these hours. However, the observed

purchasing pattern alone does not establish whether promotions or other external factors caused the increase in afternoon orders.

3. Sales Trend Analysis

The analysis of order counts by **month, year, and state** provides insight into TARGET's sales performance, growth patterns, and regional distribution of demand.

Across the analyzed period, **São Paulo, Rio de Janeiro, and Minas Gerais** consistently recorded some of the highest order volumes. These states therefore represent important markets for TARGET's Brazilian operations.

Monthly and Yearly Order Trends

Analyzing monthly order volumes across individual years helps identify potential fluctuations and recurring purchasing patterns.

The analysis shows variations in monthly order volumes across 2016, 2017, and 2018. Although some months recorded relatively high order volumes, the pattern was not consistent across all three years. For example, **October recorded a high number of orders in 2016 but a comparatively lower number in 2018**, while January showed an increase in 2018 compared with the previous year.

Therefore, based on the three-year period analyzed, there is **no sufficiently consistent pattern to conclude that TARGET experienced a stable seasonal peak or off-season period**. Order volumes appear to fluctuate from year to year rather than following a fixed seasonal pattern.

4. Order Fulfillment Efficiency

Order fulfillment efficiency was analyzed using data from the `orders` and `order_items` tables, focusing on delivery duration and the difference between actual and estimated delivery dates.

The analysis shows that the **maximum observed delivery time was 209 days**, while the minimum was **0 days**, indicating that some orders were delivered on the same day they were approved.

The comparison between actual and estimated delivery dates also reveals significant variation. In the best-performing cases, orders were delivered **9 days earlier than the estimated delivery date**, while the most delayed order was delivered **181 days after the estimated delivery date**.

This substantial variation highlights the importance of monitoring delivery performance and identifying the causes of extreme delays.

Delivery Performance by State

The analysis of average delivery time across states shows that some states achieved comparatively faster deliveries. Possible explanations include better inventory availability, demand-based inventory planning, warehouse proximity, and well-developed transportation infrastructure.

Interestingly, **São Paulo and Minas Gerais combine high order volumes with relatively low average delivery times**. Maintaining fast delivery performance in high-demand regions may be particularly important because these markets generate a substantial share of orders.

High order volumes can also create operational pressure through increased demand on warehouses, delivery personnel, vehicles, and inventory. Potential challenges include stockouts of popular products, warehouse congestion, complex multi-item orders, and limited delivery capacity.

Delivery Against Estimated Date

The analysis also identifies the top five states where average actual delivery performance was better than the estimated delivery timeline.

Possible explanations include strategically located warehouses, better inventory availability closer to customers, and the use of faster or expedited delivery services.

5. Impact on Revenue and the Economy

During the analyzed period from **2016 to 2018**, TARGET generated approximately **16 million in total revenue** in Brazil based on the available dataset.

Revenue distribution across states broadly follows the order-volume distribution. **São Paulo, Rio de Janeiro, and Minas Gerais** generated some of the highest revenue, which is consistent with their higher order volumes.

Revenue Growth Over Time

The analysis shows that **2018 generated the highest revenue**, approximately **20% higher than 2017**.

Furthermore, revenue in **2017 was approximately 99% higher than in 2016**, indicating significant year-over-year growth during the earlier stage of TARGET's Brazilian operations.

These results suggest a strong upward revenue trajectory during the period analyzed.

6. Freight Cost Analysis

Freight cost should be interpreted in relation to both the number of orders and the number of items contained within those orders.

When an order contains multiple items, the total freight cost associated with the order can increase based on the shipment requirements. Therefore, **average freight cost is not necessarily proportional to order count**.

When order volume increases, total freight expenditure may also increase, while the average freight cost per order can decrease if the additional orders are associated with more efficient shipment utilization or a different mix of order sizes.

Conversely, when order volume decreases, total freight cost may decline while average freight cost increases, potentially indicating changes in the number of items per order or shipment characteristics.

Therefore, both **total freight cost** and **average freight cost** should be analyzed together rather than using either metric independently.

7. Payment Method Analysis

The payment analysis indicates that **credit cards were the most frequently used payment method** among customers in the dataset.

One possible reason is the availability of installment-based payments, which can make higher-value purchases more affordable by spreading payments over multiple installments.

The next commonly used payment method was **UPI**, which provides a convenient digital payment option.

This suggests that maintaining flexible payment options, particularly installment-based credit card payments and convenient digital payment methods, could support customer conversion and purchasing activity.

COMPREHENSIVE BUSINESS RECOMMENDATIONS

Based on the SQL analysis, the following recommendations can help TARGET strengthen customer acquisition, increase sales, improve operational efficiency, and expand its e-commerce presence in Brazil.

1. Focus on High-Potential Regions

TARGET should prioritize **São Paulo, Rio de Janeiro, and Minas Gerais**, which consistently demonstrate high order volumes and significant revenue contribution.

Marketing campaigns, inventory planning, and fulfillment capacity can be prioritized in these regions based on observed customer demand.

2. Leverage Customer Purchasing Behavior

TARGET should analyze historical purchasing behavior to identify customer preferences, purchasing frequency, and potential customer lifetime value.

These insights can be used to develop personalized product recommendations, targeted promotions, and customer retention strategies.

3. Utilize High-Activity Hours

Since order activity is relatively high between **1 PM and 6 PM**, TARGET could test targeted promotions and personalized campaigns during this period.

A/B testing can be used to determine whether afternoon-specific campaigns actually improve conversion rates and sales.

4. Improve the Customer Journey

TARGET should provide a smooth and transparent purchasing experience by:

- Offering flexible payment options, including installment-based credit card payments.
- Providing clear order-status updates.
- Communicating estimated delivery dates accurately.
- Simplifying the checkout process.
- Improving website usability and navigation.

These improvements can help reduce friction during the purchasing process and potentially improve customer satisfaction and conversion.

5. Optimize Product Assortment and Pricing

TARGET should continuously analyze sales performance, customer behavior, and market trends to make informed decisions about product assortment and pricing.

Competitive pricing analysis can also help TARGET understand how its products compare with competing retailers in the Brazilian market.

6. Strategic Warehouse Placement and Demand Forecasting

TARGET should use historical order data and regional demand patterns to forecast future demand, particularly in high-volume markets such as **São Paulo, Rio de Janeiro, and Minas Gerais**.

Strategically positioning inventory or fulfillment facilities closer to high-demand regions could reduce delivery times and improve product availability.

7. Improve Delivery and Fulfillment Operations

TARGET should focus on reducing delivery delays by optimizing:

- Order processing
- Picking and packing
- Inventory availability
- Warehouse operations
- Delivery capacity
- Carrier selection

The large gap between estimated and actual delivery dates observed in some orders indicates an opportunity to improve delivery planning and monitoring.

8. Optimize Freight Costs

TARGET can negotiate better freight rates with carriers based on high-volume routes and explore alternative carriers or a multi-carrier strategy.

A **tiered delivery model** could also be considered:

- Lower-cost standard delivery for less urgent, lower-value orders.
- Faster express delivery for high-value or time-sensitive orders.

This approach could help balance customer expectations with transportation costs.

9. Develop Targeted Marketing Campaigns

TARGET should use customer purchase history and preferences to create personalized promotions and discounts.

Rather than relying solely on broad marketing campaigns, customer segmentation can be used to target customers based on purchasing behavior, location, product preferences, and potential lifetime value.

10. Use Data-Driven Inventory Management

Historical sales and customer demand data should be used to optimize inventory levels.

A data-driven inventory management approach can help TARGET:

- Reduce stockouts.
- Minimize excess inventory.
- Improve product availability.
- Prepare for periods of increased demand.
- Allocate inventory more effectively across regions.

11. Continuously Monitor Business Performance

TARGET should establish a continuous analytics process to monitor:

- Order volume
- Revenue
- Customer behavior
- Delivery performance
- Freight costs
- Payment preferences
- Regional performance
- Inventory requirements

Advanced analytics and visualization tools can further help management identify trends, evaluate business performance, and make data-driven decisions.

Conclusion

The SQL analysis provides a comprehensive view of TARGET's e-commerce operations in Brazil across customer behavior, geographic distribution, sales performance, fulfillment efficiency, revenue, freight costs, and payment preferences.

The analysis indicates that **São Paulo, Rio de Janeiro, and Minas Gerais are particularly important markets**, while delivery performance and inventory management represent important areas for operational improvement. Revenue also shows strong growth across the analyzed period.

By combining regional demand analysis, customer behavior insights, targeted marketing, strategic inventory placement, optimized fulfillment, flexible payment options, and continuous data-driven decision-making, TARGET can strengthen its customer relationships, improve operational efficiency, increase sales, and support sustainable growth in the Brazilian e-commerce market.